Banking Loan Dataset - SQL Queries and Output

Queries for Q1-Q15 with results generated from Banking_Loan_Dataset.xlsx (Customers, Loans, Repayments, Branches). Amounts are in rupees as stored. Results were produced by running each query exactly as shown (SQLite).

Level 1 - Basic to Intermediate

Q1. Loan portfolio summary

Total number of loans and total loan amount.

Query

```
SELECT COUNT(*)         AS total_loans,
       SUM(loan_amount) AS total_loan_amount
FROM Loans;
```

Output

total_loans	total_loan_amount
20,500	26,174,010,000

Q2. Loan type analysis

Loans and total amount per loan type, highest amount first.

Query

```
SELECT loan_type,
       COUNT(*)         AS loan_count,
       SUM(loan_amount) AS total_loan_amount
FROM Loans
GROUP BY loan_type
ORDER BY total_loan_amount DESC;
```

Output

loan_type	loan_count	total_loan_amount
Home	5,155	14,941,510,000
Business	2,082	3,509,950,000
Personal	7,171	3,131,020,000
Auto	4,083	2,998,900,000
Education	2,009	1,592,630,000

Q3. Defaulted loans

Number and total amount of defaulted loans.

Query

```
SELECT COUNT(*)         AS defaulted_loans,
       SUM(loan_amount) AS defaulted_loan_amount
FROM Loans
WHERE loan_status = 'Defaulted';
```

Output

defaulted_loans	defaulted_loan_amount
2,028	4,030,430,000

Q4. Customer income analysis

Average and maximum monthly income per income band.

Query

```
SELECT income_band,
       ROUND(AVG(monthly_income), 2) AS avg_monthly_income,
       MAX(monthly_income)           AS max_monthly_income
FROM Customers
GROUP BY income_band;
```

Output

income_band	avg_monthly_income	max_monthly_income
High (>100K)	137,980.34	300,000
Low (<30K)	23,477.70	30,000
Mid (30-60K)	44,668.72	60,000
Upper-Mid (60-100K)	76,404.96	100,000

Q5. Repayment status summary

Records, amount due and amount paid per payment status.

Query

```
SELECT payment_status,
       COUNT(*)         AS repayment_records,
       SUM(amount_due)  AS total_amount_due,
       SUM(amount_paid) AS total_amount_paid
FROM Repayments
GROUP BY payment_status;
```

Output

payment_status	repayment_records	total_amount_due	total_amount_paid
Late	29,056	731,000,795	731,000,795
Missed	233	12,000,073	0
On time	37,910	795,958,602	795,958,602
Partial	3,228	82,166,159	52,992,505

Level 2 - Business Analysis

Q6. Loan status distribution

Percentage of Active, Closed and Defaulted loans.

Query

```
SELECT
  ROUND(100.0 * SUM(CASE WHEN loan_status='Active'    THEN 1 ELSE 0 END) / COUNT(*), 2) AS active_pct,
  ROUND(100.0 * SUM(CASE WHEN loan_status='Closed'    THEN 1 ELSE 0 END) / COUNT(*), 2) AS closed_pct,
  ROUND(100.0 * SUM(CASE WHEN loan_status='Defaulted' THEN 1 ELSE 0 END) / COUNT(*), 2) AS defaulted_pct
FROM Loans;
```

Output

active_pct	closed_pct	defaulted_pct
60.62	29.48	9.89

Q7. High-value loans

Loans above the overall average loan amount (average = 1,276,780.98).

Query

```
SELECT loan_id, customer_id, loan_type, loan_amount
FROM Loans
WHERE loan_amount > (SELECT AVG(loan_amount) FROM Loans)
ORDER BY loan_amount DESC;
```

Output

loan_id	customer_id	loan_type	loan_amount
L004601	C03020	Home	5,000,000
L006482	C00534	Home	5,000,000
L011074	C09042	Home	5,000,000
L002058	C03047	Home	4,990,000
L009078	C06617	Home	4,990,000
L009372	C04391	Home	4,990,000
L010765	C06250	Home	4,990,000
L010905	C06538	Home	4,990,000
L010927	C01889	Home	4,990,000
L011051	C01200	Home	4,990,000

Showing first 10 of 7,344 rows.

Q8. Partial payment analysis

Total unpaid amount where payment_status = 'Partial'.

Query

```
SELECT SUM(amount_due - amount_paid) AS total_unpaid_amount
FROM Repayments
WHERE payment_status = 'Partial';
```

Output

total_unpaid_amount
29,173,654

Q9. Overdue analysis

Records and outstanding amount per DPD bucket.

Query

```
SELECT dpd_bucket,
       COUNT(*)                      AS repayment_records,
       SUM(amount_due - amount_paid) AS total_outstanding
FROM Repayments
GROUP BY dpd_bucket
ORDER BY total_outstanding DESC;
```

Output

dpd_bucket	repayment_records	total_outstanding
1-29	22,847	17,941,003
90+	443	12,332,294
30-59	7,609	7,884,110
60-89	1,618	3,016,320
On time	37,910	0

Q10. High-risk loan types

Loan types with more than 100 defaulted loans.

Query

```
SELECT loan_type,
       COUNT(*) AS defaulted_loan_count
FROM Loans
WHERE loan_status = 'Defaulted'
GROUP BY loan_type
HAVING COUNT(*) > 100
ORDER BY defaulted_loan_count DESC;
```

Output

loan_type	defaulted_loan_count
Personal	797
Home	486
Business	447
Auto	222

Level 3 - Joins and Advanced SQL

Q11. Customer loan details

Loan details joined with customer city and income band.

Query

```
SELECT l.loan_id,
       l.loan_type,
       l.loan_amount,
       l.customer_id,
       c.city,
       c.income_band
FROM Loans l
JOIN Customers c
  ON l.customer_id = c.customer_id;
```

Output

loan_id	loan_type	loan_amount	customer_id	city	income_band
L000001	Auto	780,000	C02611	Ahmedabad	Mid (30-60K)
L000002	Education	300,000	C05527	Indore	Upper-Mid (60-100K)
L000003	Education	620,000	C05304	Mumbai	Low (<30K)
L000004	Auto	1,470,000	C06091	Delhi	Upper-Mid (60-100K)
L000005	Personal	400,000	C09475	Kolkata	Upper-Mid (60-100K)
L000006	Home	3,310,000	C02669	Ahmedabad	Low (<30K)
L000007	Business	480,000	C04100	Chandigarh	Mid (30-60K)
L000008	Business	970,000	C05851	Lucknow	Low (<30K)
L000009	Education	480,000	C00885	Ahmedabad	Upper-Mid (60-100K)
L000010	Auto	960,000	C00745	Chandigarh	Upper-Mid (60-100K)

Showing first 10 of 20,500 rows.

Q12. Branch performance

Loans and total amount per branch (join key: branch_id).

Query

```
SELECT b.branch_id,
       b.branch_name,
       COUNT(l.loan_id)   AS loan_count,
       SUM(l.loan_amount) AS total_loan_amount
FROM Loans l
JOIN Branches b
  ON l.branch_id = b.branch_id
GROUP BY b.branch_id, b.branch_name
ORDER BY total_loan_amount DESC;
```

Output

branch_id	branch_name	loan_count	total_loan_amount
BR015	Chandigarh Branch 1	779	1,043,580,000
BR029	Indore Branch 2	770	1,023,490,000
BR028	Madurai Branch 2	749	1,007,050,000
BR026	Lucknow Branch 2	722	969,830,000
BR011	Lucknow Branch 1	727	951,540,000
BR006	Pune Branch 1	738	939,630,000
BR012	Kochi Branch 1	733	920,560,000
BR016	Chennai Branch 2	715	914,840,000
BR008	Ahmedabad Branch 1	683	913,830,000
BR005	Hyderabad Branch 1	736	910,840,000
BR018	Delhi Branch 2	671	889,700,000
BR002	Mumbai Branch 1	649	873,280,000
BR017	Mumbai Branch 2	699	866,400,000
BR024	Coimbatore Branch 2	716	866,280,000
BR013	Madurai Branch 1	695	856,930,000
BR021	Pune Branch 2	676	849,650,000
BR020	Hyderabad Branch 2	684	849,590,000
BR019	Bengaluru Branch 2	697	840,890,000
BR014	Indore Branch 1	663	838,350,000
BR022	Kolkata Branch 2	676	834,400,000
BR007	Kolkata Branch 1	701	827,310,000
BR003	Delhi Branch 1	646	826,060,000
BR027	Kochi Branch 2	664	821,970,000
BR001	Chennai Branch 1	606	815,750,000
BR010	Jaipur Branch 1	654	814,480,000
BR025	Jaipur Branch 2	621	803,760,000
BR030	Chandigarh Branch 2	640	801,150,000
BR004	Bengaluru Branch 1	632	800,230,000
BR009	Coimbatore Branch 1	601	777,860,000
BR023	Ahmedabad Branch 2	557	724,780,000

Q13. Customers with multiple loans

Customers with more than three loans.

Query

```
SELECT customer_id,
       COUNT(*) AS loan_count
FROM Loans
GROUP BY customer_id
HAVING COUNT(*) > 3
ORDER BY loan_count DESC, customer_id;
```

Output

customer_id	loan_count
C01954	9
C03592	9
C00940	8
C01656	8
C02024	8
C02761	8
C03346	8
C06766	8
C07664	8
C09095	8
C09175	8
C09396	8
C00259	7
C00510	7
C00628	7

Showing first 15 of 1,518 rows.

Distribution of the 1,518 customers: 9 loans = 2, 8 loans = 10, 7 loans = 36, 6 loans = 129, 5 loans = 396, 4 loans = 945.

Q14. Above-average loan customers

Customers whose total borrowing exceeds the per-customer average (2,991,315.43).

Query

```
WITH customer_totals AS (
    SELECT customer_id,
           SUM(loan_amount) AS total_borrowing
    FROM Loans
    GROUP BY customer_id
)
SELECT customer_id,
       total_borrowing
FROM customer_totals
WHERE total_borrowing > (SELECT AVG(total_borrowing)
                         FROM customer_totals)
ORDER BY total_borrowing DESC;
```

Output

customer_id	total_borrowing
C09095	19,700,000
C08776	16,690,000
C05032	16,140,000
C01820	15,990,000
C00905	15,610,000
C03565	14,890,000
C04590	14,680,000
C00725	14,170,000
C07309	13,940,000
C07201	13,890,000

Showing first 10 of 3,648 rows.

Averages are taken over the 8,750 customers who have at least one loan.

Q15. Rank loan types

Loan types ranked by total loan amount using a window function.

Query

```
SELECT loan_type,
       SUM(loan_amount) AS total_loan_amount,
       RANK() OVER (ORDER BY SUM(loan_amount) DESC) AS amount_rank
FROM Loans
GROUP BY loan_type
ORDER BY amount_rank;
```

Output

loan_type	total_loan_amount	amount_rank
Home	14,941,510,000	1
Business	3,509,950,000	2
Personal	3,131,020,000	3
Auto	2,998,900,000	4
Education	1,592,630,000	5